



8

Sound and Noise



All objects around us make different sounds. We can hear these sounds with the help of our ears. Some sounds are pleasant to hear, whereas other sounds may seem unpleasant and harsh. We will learn more about sound and noise in this chapter.

Learn about

- Sounds around us
- Warning sounds
- Pleasant and unpleasant sounds
- Noise pollution
- Harmful effects of noise pollution
- Ways of reducing noise pollution

► Sounds Around Us

From the moment we wake up in the morning till the time we go to sleep, we hear a lot of different sounds throughout the day. Every object around us makes a distinct¹ sound. Some of these sounds make us feel happy and relaxed, whereas others may hurt our ears if we listen to them for too long. Let us learn more about the different sounds around us.

Sounds Made by Common Objects

Different objects in our homes such as utensils in the kitchen, alarm clocks, fans, telephones, refrigerators, washing machines, and air coolers make different sounds. Some of these sounds are so low that we may not be able to hear them unless we listen closely. Other sounds are loud and can be heard from some distance away. The following are some common household sounds:

- A clock produces a soft 'tick-tock' sound when its second hand moves.
- A doorbell produces a loud musical sound so that people inside the house can know when someone is at the door.
- The hinges of an old door may make a creaking sound when the door is opened or closed.



Alarm clock

Fact File

If there is no air, you will not be able to hear any sound as it needs a medium to travel.

¹distinct: different from others

- Fans and washing machines make a continuous whirring sound.
- A tea kettle makes a low hissing sound when steam escapes from it.
- A pressure cooker makes a loud whistling sound when steam escapes from it.

Sounds Made by Living Beings

All living beings make distinct sounds. Human beings mostly use the sound of their voices when speaking to each other. When we visit a crowded market, a playground, or a railway station, we hear so many sounds. We can hear announcements being made over loudspeakers, people talking to each other, children shouting and running around, and many more such sounds. We can also produce different sounds by rubbing our palms together, clapping our hands, snapping our fingers, stomping our feet, whistling, and singing. We also produce a sound when we cough or sneeze.



A crowded street

Animals and birds also make different sounds. We can easily recognize an animal or a bird by the type of sound it makes. Some common animal and bird sounds that we are familiar with include the roaring of a tiger or a lion, chattering of monkeys, howling of wolves, screeching of parrots, chirping of sparrows, cooing of pigeons, and barking of dogs. Animals and birds may use distinct sounds to send out information about the location of food and also as warning calls to inform others of danger. Some male animals and birds also use special sounds to attract mates.



Birds chirp

When we walk through a park on a windy day, we can hear the rustling of leaves in the wind. When we walk on dry fallen leaves, they make a crackling sound under our feet.



Leaves falling from a tree

► Warning Sounds

Warning sounds are loud sounds that are used to warn people about a danger or an emergency. These warning sounds are important as they alert people to either get out of, or stay away from, a place or a situation that may put their lives in danger.

The following are some common warning sounds that are used in different situations:

Sound made by a fire alarm A fire alarm makes a loud noise to warn people inside a building about a fire. When we hear a fire alarm, we must immediately leave the building through the fire exits, even if we cannot see the fire.

Police siren A police siren makes a loud noise and warns people and vehicles on the roads to allow the police vehicles to pass through, so that the police can reach the people in danger as fast as possible.

Ambulance siren An ambulance uses its siren when it is on its way to either reach a patient, or is transferring a patient to the hospital. It makes a loud noise so that the vehicles on the road allow the ambulance to pass through, so that medical help can reach the patient as quickly as possible.

Fire engine siren A fire engine uses its siren when it is on its way to extinguish² a fire that has gone out of control. It makes a loud noise so that the vehicles on the road allow the fire engine to pass through, so that the fire can be controlled and people trapped in the fire can be helped at the earliest.



A fire engine

► Pleasant and Unpleasant Sounds

Sounds can be pleasant or unpleasant. We enjoy listening to music. It is a pleasant sound. *Any sound that makes us feel happy and relaxed can be called a pleasant sound.* The following are some examples of pleasant sounds:

- 🔊 Music from different musical instruments such as a sitar, a harmonium, or a guitar
- 🔊 Sounds made by waves on a beach
- 🔊 Rustling sound made by leaves in a gentle breeze



Guitar



Harmonium



Sitar

Musical instruments produce pleasant sounds.

²extinguish: put out

- Sound of gently falling rain
- Gentle chirping of birds

The sound of people screaming or a vehicle honking does not sound very pleasing. Such sounds are unpleasant. *Any sound that makes us feel uncomfortable, irritated, or causes a feeling of being unwell can be called an unpleasant sound.* The following are some examples of unpleasant sounds:

- Honking of vehicles on the road
- Loud explosions
- Loud sound made by aircrafts while taking off and landing
- Noise from factories and generators
- Loud music

Think and Discuss



Think about a quiet place you enjoy visiting, like a park or library. Why is it important to have quiet places in a city?



Aeroplanes make a loud sound while taking off and landing.

Questions



Fill in the blanks.

1. Some sounds hurt our if we listen to them for too long.
2. The hinges of an old door make a sound when the door is opened or closed.
3. An example of a warning sound is
4. sounds make us feel happy and relaxed.

► Noise Pollution

A loud and unpleasant sound that can cause physical and mental disturbance is called noise. Listening to loud noises for a long time can be harmful to our ears.

Loud noise disturbs our comfort levels. *Excessive and continuous noise which affects human and animal health is called noise pollution.* It causes



Noise pollution

disturbances to all living beings, especially children, elderly and sick people, and animals. Noise pollution does not allow people to sleep properly, concentrate on their work, or even talk to each other.

Some common causes of noise pollution are loud noises produced by machines in factories, from loudspeakers, televisions, radios, and electrical appliances. The

noise made by aircrafts while taking off and landing is a main cause of noise pollution for people staying near airports. Unnecessary honking of traffic on roads and bursting of crackers also cause a lot of noise pollution.



Causes of noise pollution

► Harmful Effects of Noise Pollution

Constant exposure to various sources of noise, (even noise that may not be very loud) may have several ill effects on our health. Noise pollution can cause the following problems:

- Dizziness, headaches, and earaches
- Tiredness and stress
- Loss of sleep and concentration
- Hypertension and permanent hearing loss
- Noise pollution can cause hearing loss and irritability in animals too.

Case Study



Riya has been struggling with her studies since a construction project began near her school a few months ago. The constant noise from the heavy machinery and construction activities is making it hard for her and her classmates to concentrate in class. The loud noises are causing stress and headaches as well. Riya complains about this to her parents, asking them to speak to the school administration. Her parents become worried about the long-term effects of such noise and decide to take up this issue with the school authorities.

Based on the above information, answer the following questions:

- Why is there so much noise during highway construction?
- What are the harmful effects of noise pollution on students like Riya?



► Ways of Reducing Noise Pollution

Following are some preventive measures to protect ourselves from noise pollution:

- Avoid using loudspeakers for too long and too loudly.
- Regularly check electrical appliances to reduce the noise they produce.
- Traffic should be regulated to avoid noise from vehicles. Unnecessary honking should be controlled in residential areas, school, and hospitals.
- Use heavy carpets and curtains at home; they help in absorbing the noise coming from outside.
- Plant trees in the neighbourhood. Trees and plants also absorb noise to a great extent.
- Avoid watching television and listening to radio at high volumes.

Know Your SDGs

SDG 11: Sustainable Cities and Communities

(Make cities and human settlements inclusive, safe, resilient and sustainable)

Building quieter cities is a collective effort. What ideas would you share with your neighbors to reduce noise levels and create a quieter neighborhood?

Questions



Fill in the blanks.

1. Listening to loud can be harmful to our ears.
2. Noise does not allow people to sleep properly.
3. of traffic on road causes lots of noise pollution.
4. Noise pollution causes loss of and
5. Avoid watching television and listening to radio at high

Wrap Up

- Common items in our household including dishes, clocks, doors, ceiling fans, hairdryer, telephone, and vacuum cleaner produce a variety of sounds. Living beings also produce a lot of sound.
- Warning sounds warn people of danger. These sounds are made by sirens.
- Any sound that makes us feel happy and relaxed can be called a pleasant sound. Any sound that makes us feel uncomfortable, irritated, or causes a feeling of being unwell can be called an unpleasant sound.
- A loud and unpleasant sound is called noise. It can be harmful to our ears.
- Excessive and continuous noise is called noise pollution. Noise pollution has several harmful effects on health. There are several ways of reducing noise pollution.

Exercises

SECTION I



A Choose the correct option.

1. A clock produces a
a. creaking sound b. hissing sound
c. rustling sound d. tick-tock sound
2. Which of the following produces a rustling sound?
a. Table b. Leaves c. Teakettle d. Doors
3. Which one of these produces a whirring sound?
a. Doorbell b. Piano c. Dog d. Fan
4. A fire alarm produces a
a. musical sound b. loud noise
c. hissing sound d. chattering sound
5. Which of these generally cause noise pollution?
a. Firecrackers b. Loudspeakers c. Aircraft d. All of these

B Assertion and Reasoning questions

1. **Assertion (A):** Sirens are used by emergency vehicles only.
Reason (R): Sirens are loud warning sounds.
a. Both A and R are True b. Both A and R are False
c. A is True and R is False d. A is False and R is True
2. **Assertion (A):** Planting trees reduces noise pollution.
Reason (R): Trees absorb sound waves.
a. Both A and R are True b. Both A and R are False
c. A is True and R is False d. A is False and R is True

C Give two examples of the following.

1. Warning sound
.....
2. Common object's sound
.....

SECTION II



D Short answer questions

1. How does a clock produce a soft 'tick-tock' sound?
2. When does a door produce a creaking sound?

3. Give examples to suggest that we can recognize an animal or bird by the sounds it makes.
4. What are warning sounds? Give an example.
5. What is noise? How is it different from noise pollution?

E Long answer questions

1. What are the different kinds of sound we hear throughout the day?
2. Differentiate between pleasant and unpleasant sounds giving examples.
3. What are the harmful effects of noise pollution?
4. Describe different ways of reducing noise pollution.



Picture Study

Look at the pictures and answer the questions that follow.



(i)



(ii)



(iii)



(iv)



(v)

1. Write the sounds made by the given objects/living beings.

i. ii. iii.

iv. v.

2. Mention whether they are low or loud.

i. ii. iii.

iv. v.



My Learning Corner

A Think about

Prithvi's performance in school was degrading day-by-day, after they had shifted to their new home which was located close to a tile cutter factory. Can you suggest a probable reason for it?

.....

.....

B Try out

1. Make a list of the various kinds of typical sounds that you occasionally hear in your daily life (e.g., water drop falling into a bucket, tearing paper, etc). Create a collage in your Science scrapbook using a variety of art materials to represent the different sounds. You can use paints, drawings, magazine clippings, textured materials, or found objects to create a unique composition.
2. Organize an interesting group activity/quiz to identify the pre-recorded sounds of various vehicles, musical instruments, or any other objects (using mobiles).
3. Organize a group discussion in the classroom related to the personal experiences of students regarding noise pollution, and suggesting ways to overcome them.



Self-Assessment

Now that you have completed the chapter, score each of the following tasks from 1 to 5 to indicate how well you can do them.

Score 5 = I can definitely do this.

Score 1 = I cannot do this yet.

I can...	My score
• identify objects producing pleasant and unpleasant sounds.	
• recognize sounds produced by some common objects.	
• identify sounds produced by some animals.	
• realize the importance of sound as a warning signal to save lives.	
• list the causes of noise pollution.	
• suggest some ways to reduce noise in the surroundings.	
• discuss how loud sounds affect health.	

Worksheet

Identify the type of pollution shown in the picture given below. Also, write how it can be prevented.



Type of pollution:

Definition:

.....

Preventive measures:

i.

ii.